

## Note 2.4 One-sided Limits

### One-sided Limits

- **Definition:**

If  $f(x)$  is defined on  $(c, b)$ , as  $x$  gets closer and closer to  $c(b)$  from the right(left) side but never equals  $c$ ,  $f(x)$  gets arbitrary closer to  $L$ , then  $L$  is called the the the right(left)-hand limit of  $f(x)$ . Denoted by

$$\lim_{x \rightarrow c^+} f(x) = L \quad \text{and} \quad \lim_{x \rightarrow c^-} f(x) = L$$

- **Theorem:**

$$\lim_{x \rightarrow c} f(x) = L \quad \Leftrightarrow \quad \lim_{x \rightarrow c^-} f(x) = L \quad \text{and} \quad \lim_{x \rightarrow c^+} f(x) = L$$

Corollary: If  $\lim_{x \rightarrow c^-} f(x) \neq \lim_{x \rightarrow c^+} f(x)$ , then  $\lim_{x \rightarrow c} f(x)$  doesn't exist.

- **Calculate:** Use the same laws of [note 2.2](#)

### Precise Definitions of One-sided Limits

- **Definition:**

$$\lim_{x \rightarrow c^-} f(x) = L \quad \Leftrightarrow \quad \text{for any } \epsilon > 0, \text{ there exists a } \delta > 0 \text{ such that when } c < x < c + \delta, |f(x) - L| < \epsilon$$

$$\lim_{x \rightarrow c^+} f(x) = L \quad \Leftrightarrow \quad \text{for any } \epsilon > 0, \text{ there exists a } \delta > 0 \text{ such that when } c - \delta < x < c, |f(x) - L| < \epsilon$$

### Limits Involving $(\sin \theta)/\theta$

- **Theorem:**

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1 \quad (x \text{ in radians})$$

- **Corollary:**

- $$\lim_{f(x) \rightarrow 0} \frac{\sin f(x)}{f(x)} = 1$$

- $$\lim_{x \rightarrow 0} \frac{1 - \cos x}{\frac{1}{2}x^2} = \lim_{x \rightarrow 0} \frac{\tan x}{x} = \lim_{x \rightarrow 0} \frac{\sqrt[n]{1+x} - 1}{\frac{1}{n}x} = 1$$

### Infinitesimal

- **Definition:**

If  $\lim_{x \rightarrow c} f(x) = 0$ , then we call  $f(x)$  is an infinitesimal as  $x$  approaches  $c$ .

- **Equivalent Infinitesimal:**

If  $\lim_{x \rightarrow c} f(x) = \lim_{x \rightarrow c} g(x) = 0$  and  $\lim_{x \rightarrow c} \frac{f(x)}{g(x)} = 1$ , then we call  $f(x)$  and  $g(x)$  are equivalent infinitesimal as  $x$  approaches  $c$ . Denoted by:

$$f(x) \sim g(x) (x \rightarrow c)$$

- **Corollary:**

- $\sin x \sim x$
- $1 - \cos x \sim \frac{1}{2}x^2$
- $\tan x \sim x$
- $\sqrt[n]{1+x} - 1 \sim \frac{1}{n}x$

all above are as  $x$  approaches  $c$ .

- **Theorem:**

If  $f(x) \sim t(x)$ ,  $g(x) \sim h(x)$  ( $x \rightarrow c$ ), and  $\lim_{x \rightarrow c} f(x)g(x)$ ,  $\lim_{x \rightarrow c} \frac{f(x)}{g(x)}$  exist, then

$$\lim_{x \rightarrow c} f(x)g(x) = \lim_{x \rightarrow c} t(x)h(x), \lim_{x \rightarrow c} \frac{f(x)}{g(x)} = \lim_{x \rightarrow c} \frac{t(x)}{h(x)}$$

Remarks: Only applied for quotient and product, not for sum and difference.

- **Example:**

find

$$\lim_{x \rightarrow 0} \frac{\sin x - \tan x}{(\sqrt[3]{1+x^2} - 1)(\sqrt{1+\sin x} - 1)}$$

sol: since  $\tan x \sim x$  ( $x \rightarrow 0$ ),  $\cos x - 1 \sim -\frac{1}{2}x^2$ ,  $\sqrt[3]{1+x^2} - 1 \sim \frac{1}{3}x^2$ ,  $\sqrt{1+\sin x} - 1 \sim \frac{1}{2}x$ , thus

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{\sin x - \tan x}{(\sqrt[3]{1+x^2} - 1)(\sqrt{1+\sin x} - 1)} &= \lim_{x \rightarrow 0} \frac{\tan x (\cos x - 1)}{\frac{1}{3}x^2 \cdot \frac{1}{2}x} \\ &= \lim_{x \rightarrow 0} \frac{x \left(-\frac{1}{2}x^2\right)}{\frac{1}{3}x^2 \cdot \frac{1}{2}x} \\ &= -3 \end{aligned}$$